ST344 Team Report

Team ARIMA

Data Analysis for WCivic (Phase 1)

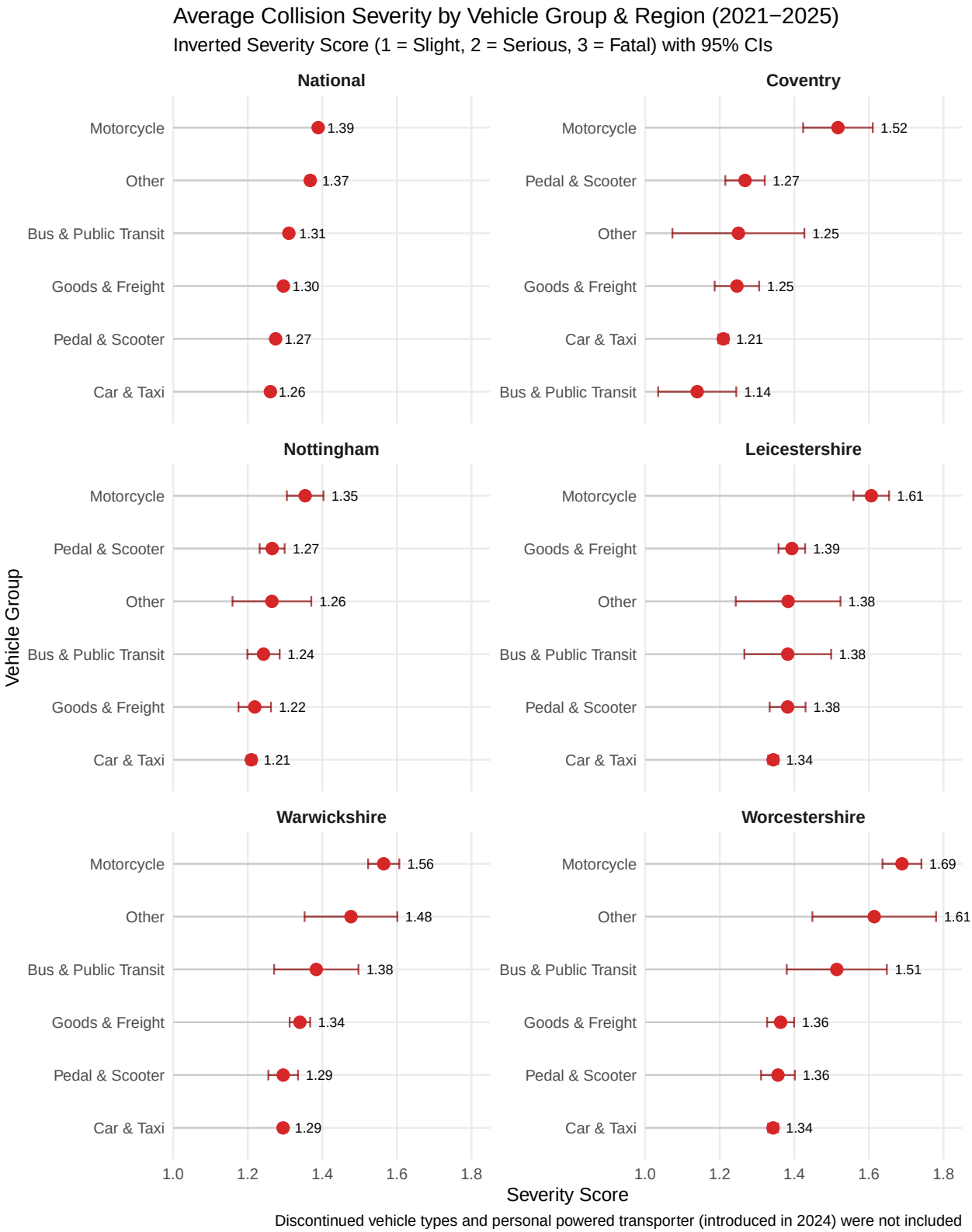
Data:

Data obtained from government website, the safety data used in this project will only include the past 5 years of data (2025-2021). Something important to note with the safety data is that collisions involving more than a single vehicle record the collision multiple times for each vehicle.

Data cleaning and preprocessing

Vehicle compositions effect on local safety outcomes

```
## `height` was translated to `width`.
```



Some 95% confidence intervals do not seem to be visable since they are very close to the mean data point. For example, the national data set has much more data so all 95% confidence intervals are very small.

A common trend across all regions in this figure is that motorcycles are on average involved in the most severe collisions. In addition, cars and taxis are on average involved in the least severe collisions for most regions. A possible reason for these trends are that motorcycles offer much less protection than that of any other vehicle type except pedal and scooter. Another reason this could occur is because most of the time motorcycles drive the fastest of any other vehicle on the road. On the other hand, cars and taxis usually have the most advanced technology and offer more safety features than any other vehicle leading to less serious collisions.

Phase 2 Analysis